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ECOLOGY

# The Dynamics of the Restoration of Mechanical Damage to Colonies of the Scleractinian Coral *Porites lutea* under Conditions of Competition with Algal Settlers for Substratum

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**Abstract**—Natural and anthropogenic catastrophes in the last decades of the 20th century and at the beginning of the new century have evoked significant changes in the biological diversity of coral reefs, viz., the mortality of hermatypic scleractinian corals that are replaced by communities of marine algae and seaweeds. In this connection, the study of the dynamics of algal colonization of a newly formed substratum and competition for the substratum between hermatypic corals and algae is of topical interest. This paper describes the results of investigations conducted in 2003 at the Sesoko Marine Biological Station of Ryukyu University (Okinawa, Japan). Colonies of the massive scleractinian coral *Porites lutea* were sampled from a fringing reef of Sesoko Island. Damages were inflicted on the upper portions of the colonies, the damaged coral fragments were placed in aquaria and maintained for six months under different light intensities in the absence of herbivorous and corallivorous animals. The experiments showed that under bright (70–90% of incident photosynthetically active radiation, $PAR_0$) and under moderate light (20–30% of $PAR_0$), the damaged parts of colonies of about 25 $cm^2$ were overgrown by newly formed polyps within six months. Under dim light (2–5% of $PAR_0$), this period was not sufficient for recovery from lesions. During the first month, the recovery rate of colonies was the highest and depended on the physiological state of the corals and light intensity: in bright and moderate light it was two times and more higher than under low light. During the subsequent months, the recovery rate was significantly lower and depended mainly on the degree of overgrowth of dead portions with algae and on their species composition. Under bright and moderate light, algal colonization of damaged colonies started with the settling of colonial and filamentous diatoms, cyanobacteria and green algae, followed by the formation of the algal turf community. Communities of red calcareous crusts and fleshy algae were mainly formed on damaged corals under low light. A comparison of experimental results showed that damaged colonies maintained in aquaria in the absence of herbivorous and corallivorous animals recovered 1.5 times faster than in the sea. We hypothesize that herbivorous fish have a negative rather than positive influence on the recovery of damaged colonies.

**Key words:** marine algae, damaged coral colonies, competition, recovery.

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The catastrophic loss of species diversity of scleractinian corals and reduction of their distribution areas on most coral reefs of the World Ocean in the last decades of the past century and at the beginning of the new millennium, which were caused by natural and anthropogenic catastrophes [4, 5, 10, 13, 16] or by settlement of marine plants on damaged reefs [7, 8, 9] made it urgently important to study the competitive relationship between seaweeds and corals which ultimately determine the fate of damaged coral reefs [1, 14, 20, 21, 23].

We have previously studied the dynamics and peculiarities of seaweed settling on mechanically damaged and dead colonies of the coral *Porites lutea* on the fringing reef of Sesoko Island (Okinawa, Japan) and investigated the pattern and the rate of regeneration of this damage [24]. It has been shown that the first settlers on damaged portions of coral colonies and on coral rolled fragments are cyanobacteria, diatomeans, and green filamentous algae. The community of algal turf widely distributed on coral reefs [11, 12, 17] forms on colony lesions and coral pebbles under conditions of sufficient illumination during 6 months. At the same time, low illumination (2–5% of incident photosynthetically active radiation, $PAR_0$) favors mainly the formation of monodominant communities of red calcareous crustose algae, brown crustose algae, and green macrophytes. The formation of these communities was affected by the grazing of herbivorous and corallivorous fish. The first feed on specific algal species of the growing communities or "cut" the plant cover, eating young parts of the thalli [2, 3], and the others, such as parrot-fishes, scrape live tissues and the skeleton surface (to 2–3 mm deep) off damaged coral colonies. Moreover, algae grown on wounds can be eaten by sea urchins, crabs and mollusks; and the young regenerating tissue of corals is subject to grazing by the gastro-

pod mollusks *Drupella* spp. or *Habromorula spinosa* (these are of wide occurrence in Okinawa) and by the sea stars *Acanthaster* spp. [15, 18, 25, 27, 28].

The formation of algal communities over the surface of damaged coral colonies in the absence of herbivorous and corallivorous animals can be traced in two ways: first, by isolating the damaged colonies from these animals with a fine net or, secondly, by placing the damaged colonies in an aquarium. But the first method does not protect corals and algae from minor animals, while the second method changes the conditions of coral existence still more drastically than the first one.

To study the formation of algal communities on damaged coral colonies and to estimate the rate of regeneration of coral polyps under conditions of different illumination and in the absence of herbivorous and corallivorous animals, we have chosen the method of maintenance of corals in a run-through aquaria. With this approach, the aquarium is supplied with natural sea water pumped from the nearby coral reef without filtration or previous settling, i.e. without significant changes of the important characteristics, such as temperature, content of nutrients or zooplankton concentration.

The aims of the study were: (1) to investigate the dynamics of algal settling on mechanically damaged colonies of the scleractinian coral *Porites lutea* under different conditions of illumination and in the absence of herbivorous and corallivorous animals; (2) to estimate the pattern, dynamics and the rate of restoration of coral polyps on damaged surfaces of the colonies; and (3) to elucidate the role of algae-settlers in restoration of colonies after mechanical damage.

## MATERIAL AND METHODS

### *Area and Time of Research*

The experiments were carried out at the Sesoko Marine Biological Station (Sesoko Island, Okinawa, Japan) in February–August of 2003. The materials for the experiment were collected on the fringing reef opposite the Sesoko Station (26°38′N, 127°52′E). The temperature of the surface water layers varied from 18 to 24°N in winter, from 22 to 28°N in spring and from 24 to 29°N in summer. The photosynthetically active radiation falling on the water surface in a mid sunny day was 800–1400 μmol/(m$^2$ s) in winter and 1100–1600 μmol/(m$^2$ s) in summer.

### *Experimental Procedures*

Specimens of massive colonies of the scleractinian coral *Porites lutea* (Edwards and Haime 1860), 0.5–1 kg in weight (figure, A), were collected in shadow sites of the coral reef at a depth of 1.5–2.0 m and maintained in an aquarium overnight. Then the colonies were divided in two halves using a hand saw, and the upper (evenly illuminated) parts were taken for the experiment. The corals colonies were injured by cutting at a right angle a portion of the skeleton with tissues to obtain both vertical and horizontal surfaces. The total area of the wound was about 25 cm$^2$ (figure, B). A total of three damaged fragments were placed in each aquarium.

The water temperature in the aquaria was 20–26°C in winter and spring in the day time and 18–24°C at night, in summer it was 28–31 and 26–28°C, correspondingly, i.e. the water temperature at night was 2–3°C lower and in the day time 1–2°C higher than in the sea. Water in the aquaria was intensively aerated; water exchange was maintained at about 30% per hour. Radiation on the sample surface was 70 to 90% $PAR_0$ in the open aquaria and from 2 to 5 and from 20 to 30% $PAR_0$ in aquaria covered with a plastic net. The rates of restoration of the colonies or coral tissue regeneration were measured during the 1st and the 3rd months of the experiment. The species variety of algae-settlers, the projective covering of the substrate with algae and the density of algal populations were analyzed after the 1st, the 3rd, and the 6th month of the experiment.

### *Measurement of the Rate of Colony Recovery*

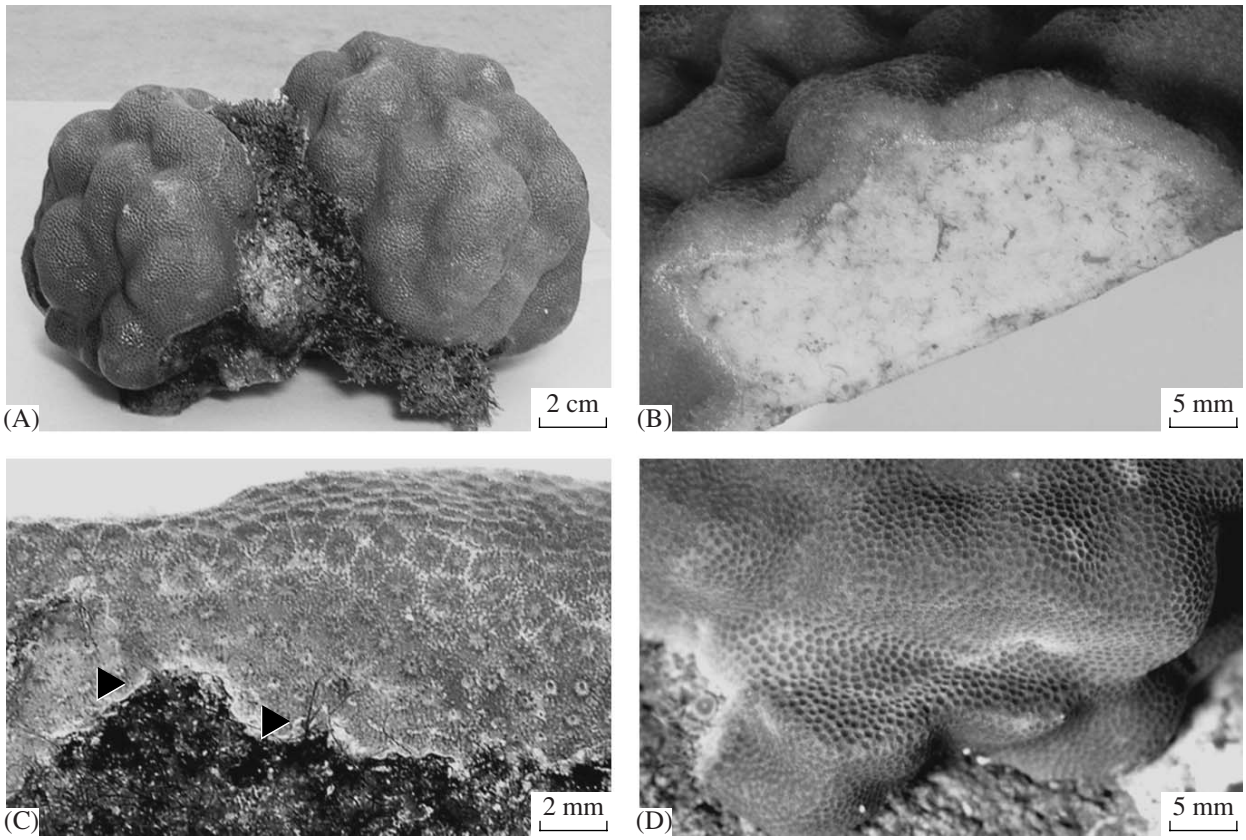
To determine the rate of recovery from damage, the lesions of coral colonies were photographed with an Olympus C5050Z digital camera immediately after injury and in 1 and 3 months afterwards. In the print, the distance was measured from the lesion center to the edge of the live coral tissue. From six to ten measurements were made and the mean rate of regeneration or the linear velocity of spreading of the healthy coral tissue over an injured surface was calculated. The mean rate and the standard deviation were calculated for three specimens ($n = 3$) based on these measurements.

### *Determination of the Projective Covering of the Lesion Surface with Algae, Analysis of the Algal Species Composition, and Calculation of the Population Density*

The damaged colonies were photographed after the 1st and the 3rd months of maintenance in the aquaria. The projective covering of the substrate with algae were determined with the use of the "paper" and "weight" methods [1]. Photographs of coral lesions were printed out on a dense thick paper of fine quality (Epson KA45OPM) with an enlargement to 3–5 times the original size. The printed images of the dead part of the wound were cut out and weighed; the algal image was then cut out of the weighed photo fragment and also weighed. The relative projective covering of the substrate surface with algae was calculated by the formula:

$$S = w_2/w_1 \times 100,$$

where $S$ is the relative projective covering of the substrate with algae, in % of the total area of the wound fragment image; $w_1$, the weight of the paper fragment




Regeneration of damaged colonies of the coral *Porites lutea*. A, one of the coral colonies used in the experiment; B, mechanically damaged colony after 1 month of exposure to moderate illumination; C, regeneration of the damaged colony after 3 months of exposure to moderate illumination in an aquarium (arrows showing algal communities hindering the healing of the lesions); D, a completely recovered colony fragment after 6 months of maintenance at moderate illumination.

with the image with wound surface; $w_2$, the weight of the paper piece with the image of algal settlers. The mean value and standard deviation were calculated on the basis of analysis of three specimens ($n = 3$).

The community of algal settlers on coral lesions was studied under a light microscope. For this, the surface of a specimen was previously examined under a stereoscopic binocular microscope and a small piece of the macroalgal thallus or the microalgal colony was cut for taxonomical identification of the species under a light microscope. The population density of specimens of the definite species within a community was determined visually at three different sites for each specimen at random. The coral surface and lesions were constantly wetted with seawater during the analysis. Each sample was examined no longer than 5 min.

### *Irradiation Measurement*

Irradiation of the water surface and the sample in aquaria (photosynthetically active radiation, PAR) was measured with the use of a LiCor (LI-192 CB) photometer at 9:00–10:00, 13:00–14:00, and 17:00–18:00 o'clock. The relative radiation (PAR) at the sample was calculated as the percentage of the $PAR_0$.

### *Statistical Analysis*

The statistical analysis was conducted with the use of the *t*-test. The differences were recognized as significant at $P < 0.05$.

## RESULTS

A total of 92 algal species were found on the fragments of damaged coral colonies. The relative content of different algal orders was as follows: 9% Bacillariophyta (B), 22% Cyanobacteria (C), 17% Chlorophyta (Ch), 13% Phaeophyta (Ph), and 39% Rhodophyta (Rh). After the first month of colonization, the projective covering of the surface of lesions with algae was from 3 to 7%. The algae colonized inhabited only the dead portions of the colony free of the coral tissue. After 3 months of maintenance of the damaged colonies in the aquarium, the projective covering of the damaged surfaces with algae increased to 40–90% (figure, C). After 6 months of the maintenance, the lesions were completely overgrown with coral polyps (figure, D) if the colony was exposed to bright and moderate light, and partly overgrown in dim light.

By the end of the 1st month of the exposure, a total of 52 algal species were found on coral wounds: 41 species in bright light, 34 species in moderate light, and only 17 species in low light. The relative number of algae (first settlers) of different orders settled on the wounds was as follows: 13% (B), 28% (C), 14% (Ch), 12% (Ph), and 33% (Rh) in bright light; 9% (B), 32% (C), 18% (Ch), 9% (Ph), and 32% (Rh) in moderate light; and 22% (B), 17% (C), 11% (Ch), 22% (Ph), and 28% (Rh) in low light. In bright and moderate light, the following species dominated or were of frequent occurrence: *Nitzschia longissima, N. closterium*, and *N. seriata* (B); *Leptolyngbya tenui* and *Calothrix* sp. (C); *Ulva clathrata* and *Derbesia marina* (Ch); *Hincksia mitchelliae* (Ph); and *Ceramium flaccidum* (Rh). By the end of the 3rd month of the experiment, the number of algal species settled on wounds at high illumination increased up to 60, of these 13% were identified as (B), 23% (C), 23% (Ch), 13% (Ph), and 28% (Rh). Under conditions of moderate illumination, 86 species were identified, of them 9% were (B), 24% (C), 16% (Ch), 13% (Ph), and 38% (Rh). At low illumination, algal species were less numerous than at bright and moderate light: from 42 algal species, 12% were (B), 17% (C), 5% (Nh), 17% (Ph), and 9% (Rh). At the same time, we failed to find algae-first settlers, such as *Lyngbya semiplena, Calothrix contarenii* (C), *Herposiphonia secunda, Spermothamnion* sp. and *Stylonema alsidii* (Rh), or *Cladophora vagabunda* and *Cladophora* sp. (Nh). The coral lesions remained to be overgrown by algae that were recorded previously as dominant and frequently occurring. As well, new dominants were recorded: *Lyngbya majuscula*, *Oscillatoria* sp., *Spirulina subsalsa* (N), *Ulva compressa* (Ch), and *Polysiphonia* sp. (Rh) at bright illumination; *Licmophora lyngbyei* (B), *L. majuscula, Oscillatoria* sp., *Tolypothrix* sp. (C), *Rhipidosiphon javensis* (Ch), *Sphacelaria novae-hollandiae, S. rigidula* (Ph), *Polysiphonia* sp. (Rh), *Diatoma* sp. (B), *Jania capillacea*, and *Gelidiopsis intricata* (Rh) at low illumination.

The highest mean rate of recovery of the coral tissue on damaged surfaces, 0.22 ± 0.06 mm per day, was recorded in the first month of maintenance of coral colonies in the aquarium at a moderate illumination of 20 to 30% $PAR_s$. In bright light (70–90% $PAR_0$), the rate was also at a high level of 0.19 ± 0.06 mm per day. At low light of 2 to 5% $PAR_0$, the mean rate of restoration of the coral tissue on damaged parts was lower than at bright and moderate light, at about 0.12 ± 0.09 mm per day. During the 2nd and the 3rd months, the rate of regeneration reduced considerably ($P < 0.05$): to 0.07 ± 0.02 mm per day in bright light, to 0.08 ± 0.01 mm per day at moderate light and to 0.05 ± 0.02 mm per day at low light.

## DISCUSSION

### *Dynamics of Algal Colonization of Mechanical Injuries of Coral Colonies*

In nature, algal overgrowth of coral colonies depends mostly on the degree of damage of coral tissue or on the physiological state of the tissue. Dead colonies (or their portions) are settled with algae immediately after damage [7, 24]. So, after the mass bleaching of corals on the Great Barrier Reef of Australia, all dead coral colonies (or colony parts) or those that lost most of zooxanthella population were colonized by algae during 1–2 months, while the less damaged colonies were restored and were not subjected to algal colonization [7, 22]. As was shown earlier [22] and in the present study, algae cannot colonize mechanically damaged soft coral tissue of corals (healthy before the injury), but immediately settle on damaged skeletons (figure, C). So, mechanically damaged live tissue regenerates very quickly (during 2–4 weeks), with settled spores or algal fragments entombed into the skeleton. According to our data, this happens in the range of illumination of 2 to 90% $PAR_0$. However, algae settle on live tissue of coral polyps weakened by high temperatures (coral bleaching) [7], by some diseases [19, 26] or negative allelopathic influences of other organisms [9, 14, 20]. Weak polyps are unable to clean the surface of coral tissue from settled particles, including algal spores and fragments. Most frequently, the weakened coral tissue is colonized by blue–green [7] or red algae, such as *Centroceras clavulatum* or *Ceramium* spp., with their rhizoids penetrating into the live coral tissue [20]. Immediately after damage to the coral, the dead portion of the wound becomes a favorable substrate for algal settlement. In this case, the settled algal spores usually attach, not to a bare substrate, but to a film formed of mucus, coral tissue remains, and organic and inorganic particles settled on its surface [6, 7, 9].

In our experiments, by the end of the 1st month of the exposure, we found 53 algal species on the dead parts of lesions, mostly with the predominance of microscopic and thin filamentous Bacillariophyta and Cyanobacteria. We did not establish any dependence of the species composition of algae-first settlers on illumination conditions. The predominance of blue–green algae among first settlers was earlier recorded on corals subjected to bleaching [7] or an impact of thin water jet or an osmotic shock [20]. After 3 months of observations, the projective covering of the dead surface of coral lesions reached 80 to 90%, while the number of algal species settled on lesions increased to 92. The filamentous and finely-branched algae common on algal turf dominate at bright and moderate illumination. At low light intensity, mostly red algae, often calcareous forms dominated, rather than turf algae.

A comparison of the species compositions of the algae overgrowing artificial mechanical damage on colonies of the coral *Porites lutea* exposed in the sea and in aquaria [24] has shown that the number of algal species in the aquaria was 1.5 times higher than that in the

sea. On some wounds of corals placed in the aquaria, we recorded algal species of rare occurrence, such as *Nitzschia longissima* (B), *Derbesia marina, Rhipidosiphon javensis* (Ch), *Sphacelaria rigidula* (Ph) and *Ceramium flaccidum* (Rh). We assume that the absence of herbivorous animals in the aquaria made it possible for these algal species to reach the maturity and to reproduce. In the sea, young sprouts of most filamentous algae are continually grazed by fish, this hinders the reproduction and distribution of the algae [2].

### *The Relationship between Algae-Settlers and Coral Polyps in Damaged Colonies*

As was shown in earlier studies [20, 21] and in the present work, mechanically damaged coral tissue recovers completely in 2–3 weeks after damage. During this period, algal spores can freely germinate and attach to the dead surface of lesions, but cannot settle on the damaged tissue, since regenerating coral tissue is able to self clean; the mechanism involves entombment of foreign objects (as well as algal settlers) into the coral skeleton [20]. At the next stage of the regeneration, when the coral tissue starts to occupy the surface of the dead portion of the lesion, the algal overgrowth of the surface becomes a physical or sometimes a chemical [20] obstacle to this expansion of the coral tissue. Consequently, the rate of the polyp tissue regeneration (or the rate of wound healing) is gradually reduced.

The competition between the settled algae and the coral polyps usually comes to an end either with complete regeneration of polyps on the dead surface of the wound (Fig., D) or with the overgrowth of the wound by a dense community of filamentous algae accompanied also by single large algal thalli or calcareous crusts. Victory in this competition for substratum between coral polyps and algal settlers depends in time mostly on the physiological state of the competitive organisms, size and toxicity of colonizers, as well as on environmental factors, such as light conditions in the habitat and activity of herbivorous and, probably, coral-livorous fish [14]. Healing its lesions, the coral tissue entombs the algae settled onto the damage into its skeleton [20]. However, we have previously shown that when the wound is inhabited by the blue–green algae *Lyngbya majuscula* and *L. semiplena*, the coral tissue ceases its advance over the wound surface, since the tissue is unable either to overgrow the algae or to entomb them into the skeleton, probably, because of the presence of toxins in the algae [24]. We have not found significant differences in the rates of regeneration of corals in the range of light intensities from 20 to 90% $PAR_0$. We hypothesize that this can be explained by the similar physiological state of coral polyps dwelling in the range of illumination, as well as by the similarity of the population density and species composition of algal communities competing with the polyps for substratum. The decrease in light intensity to 2–5% $PAR_0$ caused a sharp, nearly two–three fold reduction of the rate of regeneration of coral colonies, which may result from both a reduced level of autotrophic production of corals and a change in the species composition of algal competitors. At low light, the lesions were frequently colonized by red calcareous algae, which form a serious barrier against the advance of the coral tissue over the wound.

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